

Afterwards, the encrypted chunks are joined, preserving the order of the cleartext chunks. The payload of the last chunk may be smaller than 32KB.

Name shortening: This layer doesn't modify any file contents but it limits the path length to ensure compatibility with certain Microsoft products that do not support long paths. Even with the flattened directory structure achieved during file name encryption, the file path might be longer than 255 characters. If a ciphertext file name exceeds the threshold of 129 characters, it is replaced by its much shorter SHA-1 hash and gets a *.lng* file extension. Additionally, a similarly named metadata file is created in the 'm' directory containing a reverse-mapping back to the original name. This layer doesn't provide any additional security but the main purpose is to maximise compatibility.

Target directory: This is the final layer. After the files have been processed by all the above layers, they are finally stored into the chosen directory. This is where Cryptomator's final job is done, and lastly, the files are synchronised with the cloud storage.

Effectiveness

Cryptomator is a very useful disk encryption FOSS and it can be easily installed on many types of operating systems, running on a desktop PC or laptop, for free. It can also be used on a smartphone using an Android/iOS app and one can also back up data stored on the smartphone to the cloud. However, the Cryptomator apps for smartphones are not free to download. Cryptomator uses transparent encryption to store data on the cloud and this makes it more secure. It also becomes easy to store data online, because it's compatible with most online cloud

The pros and cons of Cryptomator

Pros	Cons
Free to use	No sharing functionality
Open source and can be modified	No contact options
Easy to use	
Can be used on smartphones also	
Community support available	
Online documentation	

storage services like Google Drive and Dropbox.

The password used for the Cryptomator vault is not shared online and remains a local password. This makes it more reliable and prevents the chances of online hacking. The encryption used by Cryptomator is also well secured, and it not only encrypts the data but also the encryption keys, for security. Cryptomator does the encryption and decryption of data, on the fly; that's why it is very effective in terms of security and performance. Moreover, as it is FOSS, it is free of any backdoor or malware issues and can be trusted as reliable software. It uses the Advanced Encryption Standard (AES) method to encrypt the data and the passphrase is protected against brute-force attacks using Scrypt – a password-based key derivation function.

Ease of use

Cryptomator is surprisingly very easy to use, unlike many open source software that are difficult to use. It can be easily downloaded from the site <https://cryptomator.org/downloads/#winDownload> and can be installed on a desktop PC or laptop for free. However, Cryptomator has the option for donation on its site, and its

smartphone apps can be downloaded from the app store by paying online. Cryptomator supports many operating systems like Windows, MacOS, Linux, Android and iOS. That's why it's very convenient to run Cryptomator on any computer or smartphone.

The user interface (UI) of Cryptomator is properly designed, with its components well organised, making it very easy to use both on a computer and a smartphone. Cryptomator also has an online support system for troubleshooting, so anyone can get help regarding any problem related to this software. The documentation on the terms of use can be found at <https://docs.cryptomator.org/en/latest/>.

Even though Cryptomator has a few drawbacks, it still beats many other disk encryption FOSS, in terms of functionality and security. Most disk encryption software store encrypted data locally on a hard drive, but Cryptomator makes it possible to store encrypted data on the cloud, easily and reliably. With the help of this disk encryption FOSS, we can store sensitive data on any cloud storage service like Google Drive and Dropbox, without any difficulty. **END** 🐧

By: Debojit Acharjee

The author is a software engineer by profession and an open source enthusiast.